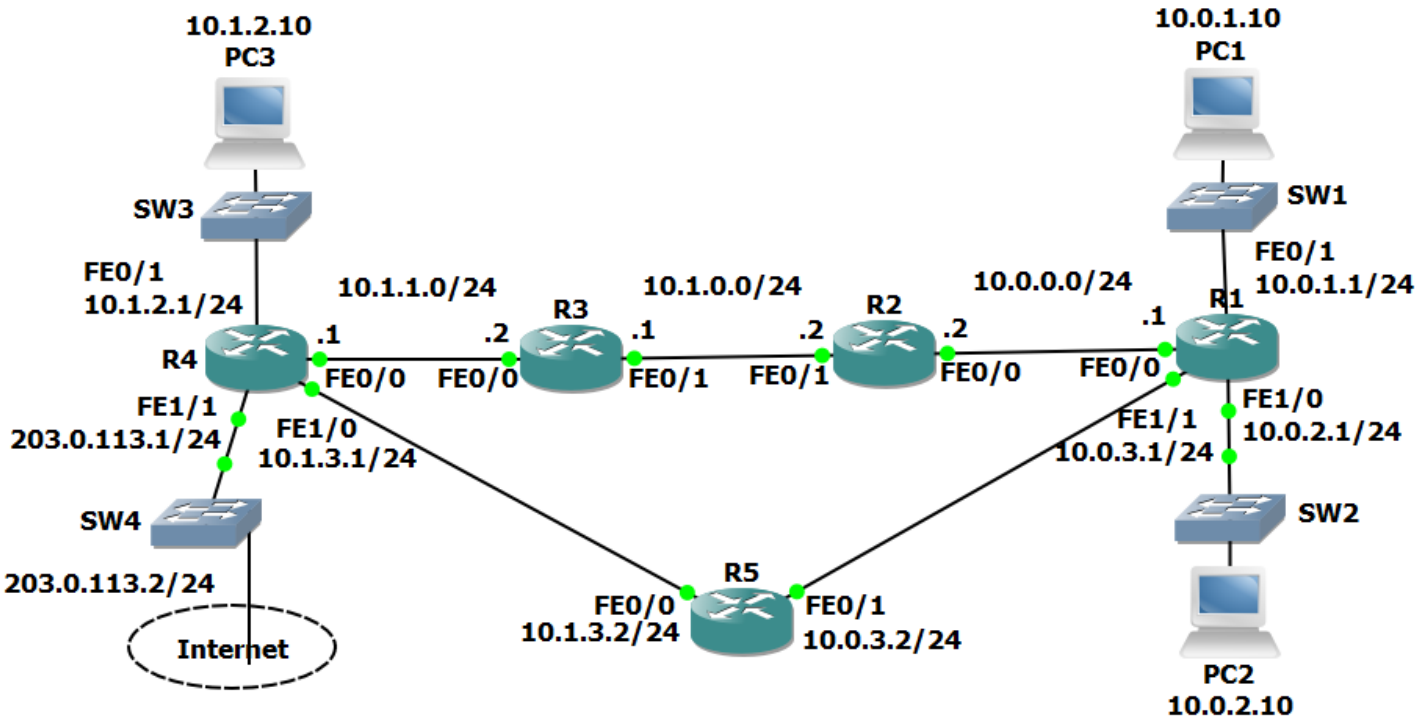
18 Connectivity Troubleshooting – Lab Exercise

In this lab you will troubleshoot a connectivity issue.

Lab Topology



Load the Startup Configurations

Download the '18 Connectivity Troubleshooting.zip' file here. *Extract* the project .pkt file then open it in Packet Tracer. Do not try to open the project from directly inside the zip file.

Troubleshoot Connectivity

1) Use ping to test connectivity from PC1 to PC3.

2) Use traceroute to determine where the problem is likely to be.

3) Determine the issue and fix it to restore connectivity between PC1 and PC3.

18 Connectivity Troubleshooting - Answer Key

In this lab you will troubleshoot a connectivity issue.

Troubleshoot Connectivity

1) Use ping to test connectivity from PC1 to PC3.

Connectivity is down.

```
C:\>ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 10.1.0.1: Destination host unreachable.
Reply from 10.1.0.1: Destination host unreachable.

Ping statistics for 10.1.2.10:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

2) Use traceroute to determine where the problem is likely to be.

Traceroute is making it as far as 10.1.0.1 on R3 so that is a good place to continue troubleshooting.

```
C:\>tracert 10.1.2.10

Tracing route to 10.1.2.10 over a maximum of 30 hops:

1 0 ms 0 ms 0 ms 10.0.1.1
2 0 ms 0 ms 0 ms 10.0.0.2
3 0 ms 0 ms 0 ms 10.1.0.1
4 1 ms * 1 ms 10.1.0.1
5 * 0 ms
Control-C
^C
```

3) Determine the issue and fix it to restore connectivity between PC1 and PC3.

The first thing to do is check that R3 has a route to PC3's network 10.1.2.0/24

```
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 9 subnets, 2 masks
S 10.0.0.0/24 [1/0] via 10.1.0.2
S 10.0.1.0/24 [1/0] via 10.1.0.2
S 10.0.2.0/24 [1/0] via 10.1.0.2
S 10.0.3.0/24 [1/0] via 10.1.0.2
C 10.1.0.0/24 is directly connected, FastEthernet0/1
L 10.1.0.1/32 is directly connected, FastEthernet0/1
C 10.1.1.0/24 is directly connected, FastEthernet0/0
L 10.1.1.2/32 is directly connected, FastEthernet0/0
S 10.1.3.0/24 [1/0] via 10.1.1.1
```

The network is not in the routing table. From the 'show ip route' command we can also see that static routes are being used. We need to add a static route for the 10.1.2.0/24 network with R4 as the next hop.

```
R3(config)#ip route 10.1.2.0 255.255.255.0 10.1.1.1
```

Lastly we need to verify connectivity is restored from PC1.

```
C:\>ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 10.1.2.10: bytes=32 time<1ms TTL=124
Reply from 10.1.2.10: bytes=32 time=1ms TTL=124

Ping statistics for 10.1.2.10:
Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
```